F1 4 A
Inline ATM mini blade fuse holder
main_fuse
SW1
KCD1 rocker switch
main_switch
VBAT 7.2V
BT1
2S 18650 pack: two 2500 mAh 20 A cells
battery
U3 · regulator
LM2596S adjustable buck module
IN+
OUT+
FB
OUT↓
5V_SERVO 5V
U5 · regulator_lights
LM2596S adjustable buck module
IN+
OUT+
FB
OUT↓
5V_LIGHTS 5V
U4 · regulator_doors
Pololu D36V50F5 5 V 5.5 A step-down regulator
VIN
EN
PG
VOUT
VBP
GND
5V_DOORS 5V
VBAT 7.2V
F2 2 A
Inline ATM mini blade fuse holder
branch_fuse_regulator
F3 2 A
Inline ATM mini blade fuse holder
branch_fuse_controller
VBAT_F3 7.2V
F4 2 A
Inline ATM mini blade fuse holder
branch_fuse_regulator_lights
D1 · eye_led_left
5 mm white LED + 100 ohm resistor
ANODE
CATHODE
D2 · eye_led_right
5 mm white LED + 100 ohm resistor
ANODE
CATHODE
A1 · controller
Arduino Nano compatible board
+5V
VBAT_F3 7.2V
5V
VIN
D3
D5
D6
D7
D8
RST
0/TX
0/RX
RST
GND
D2
D4
D11/MOSI
D12/MISO
A5/SCL
A4/SDA
D9
D10/SS
D13/SCK
3V3
AREF
A0
A1
A2
A3
A6
A7
GND
I2C_SCL_LEFT
I2C_SDA_LEFT
FACEPLATE_LEFT_PWM
FACEPLATE_RIGHT_PWM
5V_LIGHTS 5V
U1 · arc_reactor
WS2812B RGB LED ring
5V
DIN
GND
5V_LIGHTS 5V
U2 · left_repulsor
WS2812B RGB LED ring
5V
DIN
GND
5V_LIGHTS 5V
U6 · right_repulsor
WS2812B RGB LED ring
5V
DIN
GND
U7 · servo_driver_left
PCA9685 16-channel PWM servo driver board
SCL
SDA
GND
OE
V+
VCC
GND
PWM0
PWM10
PWM4
PWM1
PWM2
PWM11
PWM5
PWM3
PWM13
PWM6
PWM7
PWM8
PWM9
LATCH_LEFT_FOREARM_0_SERVO_PWM
LEFT_THIGH_SERVO_PWM
LATCH_LEFT_THIGH_0_SERVO_PWM
LEFT_SHIN_SERVO_PWM
LATCH_LEFT_SHIN_0_SERVO_PWM
LATCH_BACK_LEFT_0_SERVO_PWM
LATCH_BACK_LEFT_1_SERVO_PWM
LATCH_HELMET_LEFT_0_SERVO_PWM
LATCH_LEFT_UPPER_ARM_0_SERVO_PWM
U8 · servo_driver_right
PCA9685 16-channel PWM servo driver board
SCL
SDA
GND
OE
V+
VCC
GND
PWM3
PWM8
PWM4
PWM7
PWM5
PWM6
PWM0
PWM1
PWM9
PWM2
RIGHT_FOREARM_SERVO_PWM
LATCH_RIGHT_UPPER_ARM_0_SERVO_PWM
RIGHT_THIGH_SERVO_PWM
LATCH_RIGHT_FOREARM_0_SERVO_PWM
RIGHT_SHIN_SERVO_PWM
LATCH_RIGHT_THIGH_0_SERVO_PWM
LATCH_RIGHT_SHIN_0_SERVO_PWM
M1 · back_left_servo
DSSERVO DS3218 20 kg digital servo
Signal
M5 · helmet_left_servo
TowerPro MG90S micro servo
Signal
M21 · left_upper_arm_servo
TowerPro MG996R standard servo
Signal
M18 · left_forearm_servo
TowerPro MG996R standard servo
Signal
M2 · back_right_servo
DSSERVO DS3218 20 kg digital servo
Signal
M6 · helmet_right_servo
TowerPro MG90S micro servo
Signal
M25 · right_upper_arm_servo
TowerPro MG996R standard servo
Signal
M3 · faceplate_servo_left
TowerPro MG90S micro servo
Signal
FACEPLATE_LEFT_PWM
M4 · faceplate_servo_right
TowerPro MG90S micro servo
Signal
FACEPLATE_RIGHT_PWM
M22 · right_forearm_servo
TowerPro MG996R standard servo
Signal
RIGHT_FOREARM_SERVO_PWM
M20 · left_thigh_servo
TowerPro MG996R standard servo
Signal
LEFT_THIGH_SERVO_PWM
M24 · right_thigh_servo
TowerPro MG996R standard servo
Signal
RIGHT_THIGH_SERVO_PWM
M19 · left_shin_servo
TowerPro MG996R standard servo
Signal
LEFT_SHIN_SERVO_PWM
M23 · right_shin_servo
TowerPro MG996R standard servo
Signal
RIGHT_SHIN_SERVO_PWM
M7 · latch_back_left_0_servo
TowerPro MG90S micro servo
Signal
LATCH_BACK_LEFT_0_SERVO_PWM
M8 · latch_back_left_1_servo
TowerPro MG90S micro servo
Signal
LATCH_BACK_LEFT_1_SERVO_PWM
M9 · latch_helmet_left_0_servo
TowerPro MG90S micro servo
Signal
LATCH_HELMET_LEFT_0_SERVO_PWM
M13 · latch_left_upper_arm_0_servo
TowerPro MG90S micro servo
Signal
LATCH_LEFT_UPPER_ARM_0_SERVO_PWM
M17 · latch_right_upper_arm_0_servo
TowerPro MG90S micro servo
Signal
LATCH_RIGHT_UPPER_ARM_0_SERVO_PWM
M10 · latch_left_forearm_0_servo
TowerPro MG90S micro servo
Signal
LATCH_LEFT_FOREARM_0_SERVO_PWM
M14 · latch_right_forearm_0_servo
TowerPro MG90S micro servo
Signal
LATCH_RIGHT_FOREARM_0_SERVO_PWM
M12 · latch_left_thigh_0_servo
TowerPro MG90S micro servo
Signal
LATCH_LEFT_THIGH_0_SERVO_PWM
M16 · latch_right_thigh_0_servo
TowerPro MG90S micro servo
Signal
LATCH_RIGHT_THIGH_0_SERVO_PWM
M11 · latch_left_shin_0_servo
TowerPro MG90S micro servo
Signal
LATCH_LEFT_SHIN_0_SERVO_PWM
M15 · latch_right_shin_0_servo
TowerPro MG90S micro servo
Signal
LATCH_RIGHT_SHIN_0_SERVO_PWM

Power rails
VBAT 7.2 V from SW1: load 2.72 A typical, 7.935 A peak; source 6 A cont, 10 A peak; fuse 4 A
5V_SERVO 5 V from U3: load 0.5 A typical, 1.4 A peak; source 2 A cont, 3 A peak
5V_LIGHTS 5 V from U5: load 0.72 A typical, 2.16 A peak; source 2 A cont, 3 A peak
5V_DOORS 5 V from U4: load 2.03 A typical, 6.03 A peak; source 5.5 A cont, 8 A peak
Fuses: F1 4 A (everything on VBAT), F2 2 A (branch regulator), F3 2 A (branch controller), F4 2 A (branch regulator_lights). Branch fuses sit inline in the lead they protect.
Lever splices (star_ground, splice_vbat, splice_5v_servo, splice_5v_lights, splice_5v_doors) are junctions of their nets.
Same-net label or power symbol = same conductor. X = pin not connected.

Text to Robot
generated from design.wiring
Title:
iron_man_mark_43: schematic
47 parts, 41 nets, 119 wires, 4 rails
Size: A0
Rev: ee6e1eba
Sheet 1 / 1
File: schematic.svg
KiCad symbol style
Id: 1/1